

Question A — The Competing Race Between Repair and Detection

Alexandre Boyer

We formalize and bound $P(\tau_{\text{ARF}} < \tau_{\text{det}})$ without any assumption on the dependence between the two clocks.

Notation.

τ^*	the drift instant, known by construction
τ_{ARF}	delay until the first tree is replaced, $\min_i \tau_i$ over the M trees
τ_{det}	delay until the first external alarm, possibly $+\infty$
F_A, F_D	cumulative distribution functions of τ_{ARF} and τ_{det}
H	post-drift observation horizon, $H = 2000$
Miss	the “blind spot” event, defined in Section 1
Both delays are counted from τ^* and take integer values, a fact the bounds of Section 3 rely on.	

1 Defining the Miss Event

Everything that follows is a statement about P_{miss} , and none of it means anything until Miss is unambiguous. The natural reading — the forest repairs itself before the alarm sounds — leaves three cases open: the two clocks coincide, the detector never fires, or neither mechanism ever acts. We settle them by one definition rather than three rules.

1. A strict inequality. Both clocks count integer steps, so they can land on the same one and $P(\tau_{\text{ARF}} = \tau_{\text{det}})$ can be strictly positive: the choice is not cosmetic, and no continuity argument disposes of it — real-valued times can coincide too, all the more so for two functionals of the same error stream. It must therefore be settled on physical grounds. A tie means the alarm sounds *exactly* when the forest repairs itself: the operator is informed at that instant and nothing is lost, unlike the blind spot proper where the alarm comes late or never. A tie is a detection success, whence

$$\text{Miss} := \{\tau_{\text{ARF}} < \tau_{\text{det}}\} \quad \text{on the extended reals, strict inequality.}$$

The convention carries almost no weight in practice: ties occur 0 times out of 1906 observed races at $\lambda = 8$, 6 out of 778 at $\lambda = 25$, 0 out of 1 at $\lambda = 50$, so the non-strict reading would move P_{miss} by at most 0.003. Section 4 returns to why they are this rare.

2. A silent repair is a miss. On $\{\tau_{\text{ARF}} < +\infty, \tau_{\text{det}} = +\infty\}$ the forest repaired itself and the alarm never sounded — the blind spot in its purest form, worse than $\tau_{\text{ARF}} < \tau_{\text{det}} < +\infty$ where the operator is at least informed late. A finite τ_{ARF} is smaller than $+\infty$, so the event falls inside Miss on its own.

It must. Excluding it would restrict the count to runs where the detector eventually spoke — the very selection bias this question holds against the manuscript, since those discarded are the runs where it failed most completely. At $\lambda = 50$, where 1999 runs out of 2000 never fire, the rate would be computed over a single run.

3. A double starvation is not. On $\{\tau_{\text{ARF}} = +\infty, \tau_{\text{det}} = +\infty\}$ neither mechanism ever acted. The paradox rests on a *silent repair* erasing the evidence; if the repair never happened there is nothing to erase, and the model simply stays broken. Since $\infty < \infty$ is false, the event is excluded on its own.

Adopted convention.

$$\text{Miss} := \{\tau_{\text{ARF}} < \tau_{\text{det}}\} \quad (\text{strict, extended reals})$$

A tie goes to the detector, a silent repair is a miss, a double starvation is not — all three from the definition, none from a rule of its own. That is what makes it trustworthy rather than fitted to the answer.

2 Making the Miss Probability Observable

$\text{Miss} = \{\tau_{\text{ARF}} < \tau_{\text{det}}\}$ is defined on the complete race, with no time limit. The campaign observes $H = 2000$ steps. On runs where neither clock has fired by then, the winner is simply not known, and P_{miss} depends on what was never seen: it is not, as it stands, computable from the data.

Two ways out are available and both are unacceptable. Assuming that censored runs behave like the others cannot be checked. Discarding them reintroduces the selection bias of Section 1, and they are exactly the runs the phenomenon is about. The aim here is therefore to replace P_{miss} by the probability of an event that *is* observable — or, failing an exact rewriting, to trap it between two such probabilities.

1. What the horizon lets us see. Absolute censoring is $C_H := \{\min(\tau_{\text{ARF}}, \tau_{\text{det}}) > H\}$: neither event has happened by the horizon, so the race is undecided. Conversely $\{\tau_{\text{ARF}} \leq H\}$ and $\{\tau_{\text{det}} \leq H\}$ are observable, and sorting runs by these two gives a partition in which the verdict is known everywhere except one cell.

Region	τ_{ARF} observed?	τ_{det} observed?	Verdict
1	$\leq H$	$\leq H$	Miss decided directly (direct comparison)
2	$\leq H$	$> H$	Certain Miss — $\tau_{\text{ARF}} \leq H < \tau_{\text{det}}$
3	$> H$	$\leq H$	Never a Miss — $\tau_{\text{det}} \leq H < \tau_{\text{ARF}}$
4	$> H$	$> H$	Unknown — this is C_H

Regions 1 and 2 together are exactly $\{\tau_{\text{ARF}} \leq H\} \cap \text{Miss}$, a probability computable from the data with no assumption about what happens beyond H . Region 3 is never a Miss. Only region 4 is undetermined.

2. Trapping the unknown between two computable numbers. Since region 4 is the only undetermined one, it can shift the total by its own mass and no more. Counting it as never a Miss gives the lower bound, counting it entirely as Miss gives the upper one:

$$\underbrace{P(\{\tau_{\text{ARF}} \leq H\} \cap \text{Miss})}_{\text{computable, no assumption}} \leq P_{\text{miss}} \leq P(\{\tau_{\text{ARF}} \leq H\} \cap \text{Miss}) + P(C_H)$$

This is a Manski-type bound: partial identification. Nothing is assumed about the unobserved runs — their possible influence is simply bounded by how many there are. The width of the interval is exactly $P(C_H)$, which is to say exactly the extent of our ignorance.

3. On this campaign, the interval collapses. Section 4 measures $P(C_H) = 0$: the forest, running at maximal reactivity ($c_{\text{int}} = 1$), replaces a tree before the horizon on every one of the 2000 runs — the largest τ_{ARF} observed is 1293 steps. The minimum of the two clocks is therefore always finite, the race always decided, and the two bounds coincide:

$$P_{\text{miss}} = P(\{\tau_{\text{ARF}} \leq H\} \cap \text{Miss}).$$

Result. The unconditional miss probability equals the probability of an event observable within the horizon. This is an equality, not an assumption: the naive computation would have returned the same number, but nothing would have told us it was right. What the construction adds is the guarantee — and the condition under which it would fail. Were the forest able to stay inert past H , P_{miss} would cease to be identified and only the interval above would survive.

3 Universal Fréchet Bound

We return to the case with no finite horizon: from the marginal laws F_A and F_D alone, without any dependence assumption, we bound P_{miss} .

Lower bound. For a fixed integer s , the event $\{\tau_{\text{ARF}} \leq s < \tau_{\text{det}}\}$ always implies $\tau_{\text{ARF}} < \tau_{\text{det}}$ — it is therefore a subset of Miss, for any s . Write $A = \{\tau_{\text{ARF}} \leq s\}$ and $B = \{\tau_{\text{det}} > s\}$, so that $P(A) = F_A(s)$ and $P(B) = 1 - F_D(s)$. Fréchet’s inequality for an intersection follows from $P(A \cup B) \leq 1$:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1,$$

and since a probability is non-negative, $P(A \cap B) \geq \max(0, P(A) + P(B) - 1)$. Substituting the two marginals, $P(A) + P(B) - 1 = F_A(s) + (1 - F_D(s)) - 1 = F_A(s) - F_D(s)$, hence

$$P(\tau_{\text{ARF}} \leq s, \tau_{\text{det}} > s) \geq \max(0, F_A(s) - F_D(s)),$$

an inequality that uses the two marginal laws only, and assumes nothing about how the two clocks are related. and since the inclusion holds for every s , we optimize by taking the largest value:

$$P_{\text{miss}} \geq \max\left(0, \sup_{s \in \mathbb{Z}} [F_A(s) - F_D(s)]\right).$$

Upper bound. Take a run in Miss, so $\tau_{\text{ARF}} < \tau_{\text{det}}$, and split on where τ_{ARF} falls. If $\tau_{\text{ARF}} \leq s$, the run already belongs to the first set below, and nothing more is needed. If $\tau_{\text{ARF}} \geq s + 1$, then $\tau_{\text{det}} > \tau_{\text{ARF}} \geq s + 1$ gives $\tau_{\text{det}} > s + 1$, and the run belongs to the second. Every run in Miss therefore lies in one of the two:

$$\text{Miss} \subseteq \{\tau_{\text{ARF}} \leq s\} \cup \{\tau_{\text{det}} > s + 1\}.$$

Boole’s subadditivity, followed by optimizing over s (and the trivial truncation at 1, since a probability never exceeds 1), gives

$$P_{\text{miss}} \leq \min\left(1, \inf_{s \in \mathbb{Z}} [F_A(s) + 1 - F_D(s + 1)]\right).$$

Result — final bound.

$$\max\left(0, \sup_s [F_A(s) - F_D(s)]\right) \leq P_{\text{miss}} \leq \min\left(1, \inf_s [F_A(s) + 1 - F_D(s + 1)]\right)$$

Valid for any dependence structure between τ_{ARF} and τ_{det} — which is what makes it a *universal* bound.

Verification on a toy case. Support $\{1, 2\}$, uniform marginals: $F_A(1) = F_D(1) = \frac{1}{2}$, $F_A(2) = F_D(2) = 1$.

Coupling	Description	$P(\text{Miss})$	Position
Comonotone	$\tau_{\text{ARF}} = \tau_{\text{det}}$ always	0	= lower bound
Independent	independent draws	0.25	strictly interior
Anti-monotone	one equals 1 when the other equals 2	0.5	= upper bound

Quick check: $\sup_s [F_A(s) - F_D(s)] = 0$ (identical marginals) gives a lower bound of 0, attained by the comonotone coupling. And $\inf_s [F_A(s) + 1 - F_D(s + 1)] = 0.5$ (attained at $s = 0$ and $s = 1$) gives an upper bound of 0.5, attained by the anti-monotone coupling. The formula holds, and both bounds are attained exactly: the bound is the tightest possible on this toy case.

The bound on the estimated marginals. Evaluating the two formulas on $\hat{F}_A$ and $\hat{F}_D$, amplitude by amplitude (the 100 seeds of a cell are i.i.d., which pooling across amplitudes would not be), gives the following. Within a cell, $\hat{F}_A(s)$ is the share of the 100 runs with $\tau_{\text{ARF}} \leq s$ and $\hat{F}_D(s)$ the share with $\tau_{\text{det}} \leq s$ — two staircases rising by 1/100 at each observed value. $\hat{F}_A$ reaches 1, since τ_{ARF} is never censored; $\hat{F}_D$ does not, because a censored τ_{det} enters it at no finite s . It is therefore *sub-stochastic*, its plateau sitting at one minus the cell’s censoring rate, with the missing mass at $+\infty$ — which is how censored runs enter the bound as detector losses rather than as data to be filled in. All three columns below are *estimates* on 100 seeds — the two bounds because they are computed from $\hat{F}_A$ and $\hat{F}_D$, the middle one because it is the empirical frequency $\hat{P}_{\text{miss}}$ of the miss event, not the probability itself.

Δe	$\lambda = 8$			$\lambda = 25$		
	lower	$\hat{P}_{\text{miss}}$	upper	lower	$\hat{P}_{\text{miss}}$	upper
0.028	0.91	1.00	1.00	1.00	1.00	1.00
0.085	0.20	0.38	0.54	0.86	0.95	1.00
0.141	0.10	0.12	0.26	0.39	0.77	1.00
0.194	0.05	0.06	0.23	0.49	0.82	1.00
0.243	0.03	0.03	0.23	0.71	0.94	1.00
0.361	0.02	0.02	0.25	0.92	1.00	1.00
0.498	0.01	0.01	0.18	1.00	1.00	1.00

Three readings. First, the bound holds on *all* 60 cells of the grid (20 amplitudes $\times$ three thresholds): P_{miss} lands inside 20/20 times at each λ . This is the verification of Section 3 on real marginals, the toy case of the previous paragraph having only established that the formula is tight in the best case.

Second, the bound is not vacuous, which is worth stating because universal bounds usually are. The mean width per amplitude is 0.195 at $\lambda = 8$ and 0.106 at $\lambda = 25$; at $\lambda = 50$ it is 0.001, i.e. the bound is numerically exact. The reason is instructive: at $\lambda = 50$ the detector almost never alarms, $\hat{F}_D$ is flat at 0, and both formulas degenerate to $P_{\text{miss}} = 1$ on their own. The universal bound is sharpest exactly where the blind spot is total.

Third, and most usefully, at $\Delta e = 0.028$ and $\lambda = 8$ the estimated *lower* bound is 0.91. Being computed from 100 seeds it carries sampling error of its own, and the bound is a supremum of a difference of empirical distribution functions rather than a simple proportion, so a bootstrap over the seeds is the right instrument: it gives [0.87; 0.97], hence $P_{\text{miss}} \geq 0.88$ at the 95% level. Even at that reduced figure the statement holds *without any assumption whatsoever* about how the two clocks depend on each other — from the marginal laws alone. At the bottom of the grid, the blind spot is not an artefact of an independence hypothesis: it is a consequence of the marginals.

Why amplitude by amplitude. Applying the same two formulas to the 2,000 runs pooled gives [0.048; 0.485] at $\lambda = 8$ — a width of 0.437, against 0.195 on average per amplitude. Pooling mixes cells whose difficulty differs by two orders of magnitude, and the marginals of a mixture are compatible with far more joint behaviours than the marginals of any one cell. The loss is not an artefact of the bound but a real loss of information, and it is the same caution that Q D.1 attaches to its grid totals.

4 The Race, Measured

The apparatus above is now read on the instrumented campaign of Q C/D: 20 amplitudes $\times$ 100 seeds = 2,000 runs, $H = 2,000$. Everything in this section comes from `QCD_indicateurs_full` and is reproduced by `resultats_R2/scripts/chiffres_QA.py`.

One measurement governs the rest. τ_{ARF} is *never* censored: 0 runs out of 2,000, the largest value observed being 1,293 steps, well inside the horizon. The forest always repairs itself before H . Three consequences follow immediately, and they are the answers to the questions Sections 1 and 2 left open.

Double starvation never occurs. The event $\{\tau_{\text{ARF}} = +\infty, \tau_{\text{det}} = +\infty\}$ has empirical frequency 0 at all three thresholds. Its treatment in Section 1 remains necessary — nothing in the model forbids it, and a campaign with a shorter horizon or a slower forest would produce it — but on this grid the convention is never actually exercised.

The identification problem does not bind here. Since $\{\tau_{\text{ARF}} > H\}$ is empty, regions 3 and 4 of the partition are both empty, and in particular $C_H = \emptyset$. The width of the Manski bound, which Section 2 shows to be exactly $P(C_H)$, is therefore 0: on this campaign P_{miss} is not partially but *point* identified. The bound is not vacuous, it is simply not binding — a fact about this experimental design, not about the method.

λ	region 1	region 2	C_H	P_{miss}	Wilson 95%
8	1,906	94	0	0.0905	[0.0787; 0.1039]
25	778	1,222	0	0.9715	[0.9633; 0.9779]
50	1	1,999	0	1.0000	[0.9981; 1.0000]

Region 1 is decided by direct comparison, region 2 is a certain Miss ($\tau_{\text{ARF}} \leq H < \tau_{\text{det}}$), and regions 3 and 4 are empty. The $\lambda = 8$ value cross-checks against Q D.1, which counts the complementary event and reports $\tau_{\text{det}} < \tau_{\text{ARF}}$ in 1,819 of 2,000 runs: the two counts sum to 2,000 exactly, with no run left over — which is also how one sees that ties are absent at this threshold.

Ties are rare, but not for the reason one might assume. Over the runs where both clocks are observed, $\tau_{\text{ARF}} = \tau_{\text{det}}$ occurs 0 times at $\lambda = 8$, 6 times out of 778 at $\lambda = 25$ (0.77% of them, 0.30% of the campaign), and 0 times at $\lambda = 50$. Switching to the non-strict convention would move P_{miss} by at most 0.003, and only at $\lambda = 25$. So the choice made in Section 1 is immaterial *in fact* — but not because of any underlying continuity: both clocks are integer step counts and could easily coincide. They rarely do because the two mechanisms operate on visibly different time scales, which is the substance of the paradox rather than a technicality.

The aggregate is not the phenomenon. These three totals pool cells of very unequal difficulty and should not be quoted alone. At $\lambda = 8$, P_{miss} runs from 0.01 at the top of the grid to 1.00 at $\Delta e = 0.028$, where the detector alarms in only 10 of 100 runs within the horizon, each time after the first replacement: the aggregate 9.05% is carried almost entirely by that single weakest amplitude. Table 2 and Figure 2 of Q D.1 give the per-cell breakdown, with a Wilson interval and a censoring rate for each cell.

5 Outlook

The three bounds are established and evaluated, and P_{miss} is point identified on this campaign (Section 4), so what the Fréchet interval now measures is what the *marginals alone* concede — between 0.09 and 0.34 of probability mass at $\lambda = 8$, depending on the amplitude. Closing that residual gap is not a matter of more data: it requires knowing the *structure* of the dependence, which is precisely the object of Question B, where independence conditional on the stream yields a sharper law than any knowledge of the marginals can. The natural continuation is therefore to ask how much of this gap the conditional factorization of Q B.1 actually recovers — not attempted here.

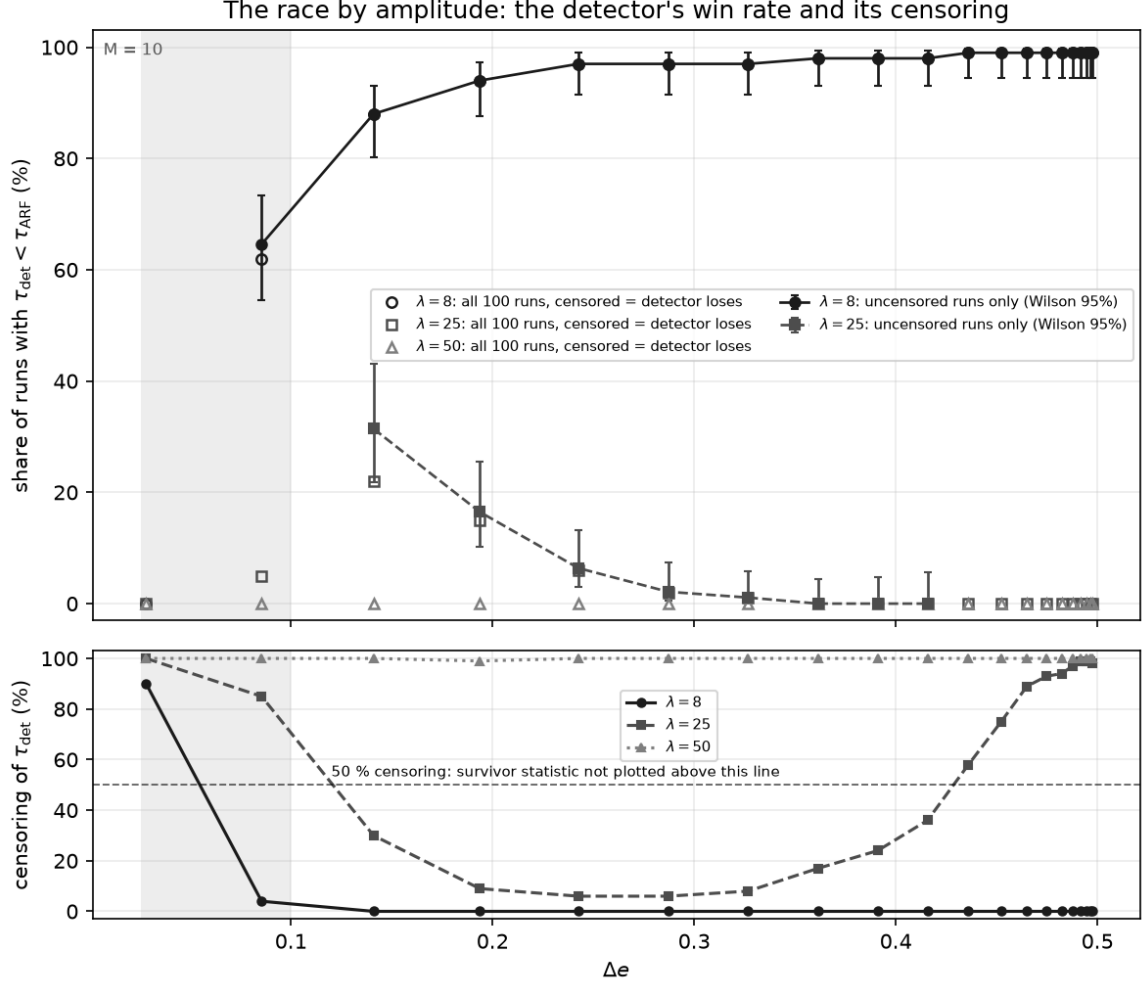


Figure 1: The race as a function of $(\lambda, \Delta e)$, the two quantities P_{miss} depends on. Top: share of runs in which $\tau_{\text{det}} < \tau_{\text{ARF}}$ — the complement of Miss up to ties — by amplitude, for $\lambda \in \{8, 25, 50\}$. Filled markers with error bars: uncensored runs only, Wilson 95% interval, drawn only where the cell's censoring is at most 50%. Hollow markers: all 100 runs, censored runs counted as detector losses, i.e. as region-2 Misses. Bottom: censoring of τ_{det} in the same cells, with the 50% line. Produced by `figures_revision_QCD.py` for Q D.1.